MODULE 4

NATURE - BIO INSPIRED MATERIALS & MECHANISMS

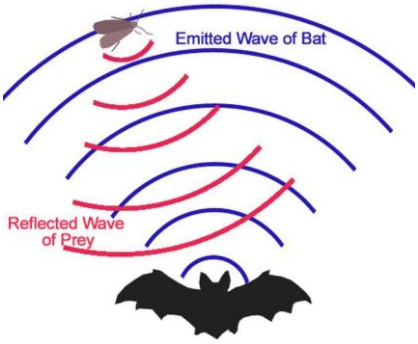
ECHOLOCATION:

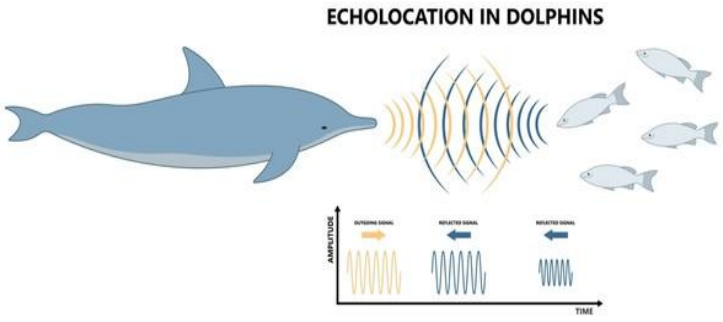
Echolocation is a biological or technological process that involves emitting sound waves and listening to the echoes that bounce back off of objects in the environment to determine their location, distance, and shape.

1. **The Mechanism:** A physiological process where an animal emits sound waves that bounce off objects, returning as echoes that provide data on distance and size.
2. **Biological Success:** Used by over a thousand species, including bats, toothed whales, and nocturnal birds to navigate and hunt in zero-light environments.

Biological Echolocation

1. Found in various animals such as bats, dolphins, and some species of whales.
2. Relies on the emission of sound waves, usually in the form of clicks or vocalizations.
3. Animals emit sound waves and listen for the echoes produced when the sound waves bounce off objects in their environment.
4. By analyzing the echoes, animals can determine the location, distance, and even the shape of objects around them.
5. This ability is mainly used for navigation, hunting, and communication in the animal kingdom.
6. Biological echolocation is a natural adaptation that has evolved over millions of years.





Technological Echolocation

1. Replicates the concept of biological echolocation using technological devices.
2. Utilizes sound waves, typically generated by artificial sources such as sonar or ultrasonic sensors.
3. These devices emit sound waves and analyze the echoes that bounce back from objects.
4. The information from the echoes is processed and interpreted by the technology to generate useful data, such as distance, location, and object recognition.

5. Technological echolocation has applications in various fields, including navigation, robotics, obstacle detection, and medical imaging.

6. It is a human-engineered solution inspired by the natural abilities of animals.

Steps involved in Echolocation:

1. **Emitting sound waves:** Animals emit high frequency (>20000Hz) sound waves.
2. **Sound wave reflection:** sound waves hit the object & reflect back, creating echo.
3. **Reception of echo:** By listening echo, animals get an information about location, size & shape of object.
4. **Interpretation of echo:** Animals brain processes info & create an image of object location & characterization.

Applications:

1. **Devices for The Blind People:** It emits sounds & listen echoes to detect obstacles.
2. **Medical imaging:** To image internal organs or tissues. Ex: fetal development, tumors detection, cardiography.
3. **Sound Navigation and Ranging:** SONAR was created to aid in navigation, detection, and communication beneath the surface of the sea. Used to find the depth of sea or distance of submarines, water hills, valleys, icebergs and sunken ships.
4. **Robotics:** To detect obstacles & navigation.
5. **Conservation:** Echolocation is used in conservation efforts to study & protect animal population. To track the movements & behavior of animals.

ULTRASONOGRAPHY:

1. Also called ultrasound imaging uses echolocation to image internal organs or tissues.
2. It is a non-invasive diagnostic technique used to image inside the body.
3. This technique is used in obstetrics, gynecology, cardiology
4. Ultrasound probes, called transducers, produce sound waves that have frequencies above the threshold of human hearing.

Instrumentation:

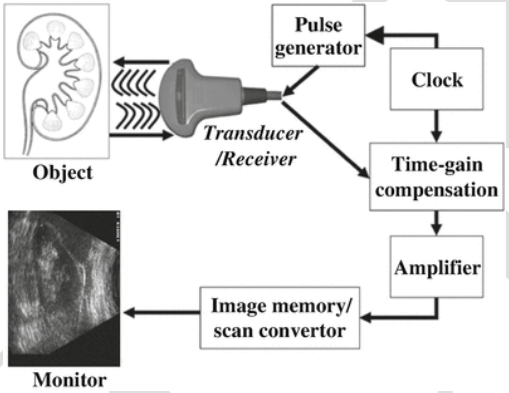
1. **Transducer:** which can both emit ultrasound waves, as well as detect the ultrasound echoes reflected back.
2. In most cases, the active elements in ultrasound transducers are made of special ceramic crystal materials called piezoelectric.
3. These materials are able to produce sound waves when an electric field is applied to them.
4. **Pulser & receiver:** The pulser circuit generates electrical impulses that are applied to a transducer causing the transducer to emit an ultrasound pulse.
5. **Receivers** detect the echo's & converts them back into electrical signals.

6. **Signal processor:** It amplifies & filters electrical signals by transducer & converts into format that can be displayed on monitor.

7. **Display:** Shows image in real time.

Working of Ultrasonography:

1. **Sound waves generation:** The transducers emit high frequency sound waves into the body.
2. **Sound waves reflection:** The sound waves bounce off the internal organs and tissue, creating echoes that are detected by the transducers.
3. **Reception of echoes:** The transducer detects the echoes and sends them to a signal processor.
4. **Image creation:** The signal processor uses the information from the echoes to create a 2D or 3D image of the internal organ, which is displayed on the monitor.



Procedure:

1. During sonography, the doctor will place a transducer (probe) directly on the skin or inside a body opening (vagina or rectum).
2. A thin layer of gel is applied to the skin directly above the organ to be scanned. Ultrasound waves are transmitted from the transducer through the gel into the body.
3. These short bursts of sound energy hit the desired organs and return to the probe as an echo. The probe diverts them to the system.
4. The system analyses data & creates an organ image.
5. The image formed depends upon the echoes received and time taken for the sound to be reflected.
6. In a diseased organ, the echoes reflected are different compared with those in a healthy organ. Therefore, the image formed is different.

Advantages:

1. Non-invasive technique yet a highly accurate scan to view insides of the body.
2. Ultrasound images are captured in real time. They can show the movement of the body's internal organs, foetus, and blood flow through the blood vessels.
3. Unlike X-ray imaging, there is no ionizing radiation exposure associated with ultrasound imaging. Hence, it can be used safely in pregnancy.

Disadvantages:

1. A USG cannot be used in organs with large amount of air, for example, for the lungs or a gassy intestine.
2. Sound waves don't travel well through bone also, so ultrasound isn't effective at imaging body parts that are hidden by bone, such as the head.
3. Ultrasound may also be unable to see objects that are located very deep in the human body.

SONAR:

Sonar, which is an acronym for "sound navigation and ranging" is a direction application of nature's echolocation technique that uses sound propagation in water used most commonly in submarines to navigate, communicate with or detect other vessels.

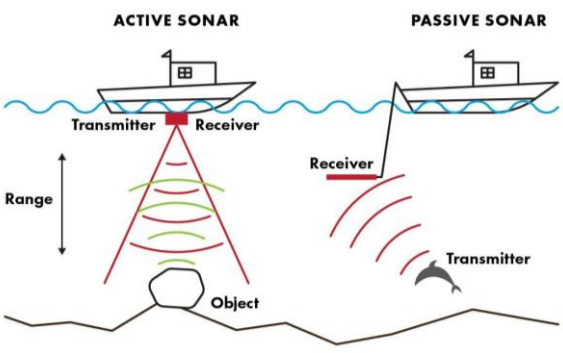
Instrumentation of sonar:

1. **Transducer:** It emits and receives sound waves. The transducer converts the electrical energy in to sound waves and vice versa.
2. **Receiver or Hydrophone:** Used to detect and measure under water sound waves.
3. **Signal processor:** It processes the signals received from the hydrophones or transducer. It includes various filters, amplifiers to enhance the signal and remove unwanted noise.
4. **Display:** This is the output component that presents the results of the signal processing. It gives the visual image or other data depending on the application.
5. **Control unit:** Used to control the settings and parameters of the system. It may include various knobs, buttons and input devices.
6. **Power supply:** Provides power to the sonar system. It may be using batteries or connected to external power source.

Types of sonar:

a. **Active Sonar:** In this, a pulse of sound is sent inside the water that hits the object and is reflected back. The distance is measured by the time which is taken by the waves to return back.

b. **Passive Sonar:** The applications of this SONAR type are used mainly in receiving sounds of the water body to locate other ships and submarines.



Applications

1. In the **fishing industries**, sonars emit signals and measure the strength of the reflected signals. The strength of the reflected signal (echo) is directly proportional to the size of the fish. This helps detect the size of fish and how they interact with the coral reefs in oceans.
2. SONAR is used for the purpose of **locating underwater hazards** and mines for the purpose of achieving safe navigation.
3. It is used for the purpose of **seafloor imaging and seafloor mapping.**
4. **Military:** For submarine detection & tracking, mine detection
5. **Environmental monitoring:** Used to monitor under water temperature, pressure and currents.
6. **Research:** Used to study underwater currents, geological formation, and marine life.
7. **Industrial application:** Used in underwater welding and drilling and in oil and gas exploration.
8. **Search and rescue:** Used to locate submerged objects like downed airplanes and sunken ships.

Advantages of Sonar Technology

1. **Versatility:** Sonar technology is versatile and can be used in a variety of applications, such as underwater navigation, mapping, and imaging, as well as for military and scientific purposes.
2. **Cost-effective:** Compared to other underwater imaging technologies, sonar is relatively cost-effective and affordable.
3. **Non-invasive:** Unlike other imaging technologies, such as diving and remote-operated vehicles, sonar does not physically disturb the underwater environment, making it an ideal choice for environmental monitoring and scientific research.
4. **Real-time imaging:** Sonar provides real-time imaging, allowing operators to quickly and easily assess the underwater environment.
5. **High resolution:** Modern sonar systems have high-resolution capabilities, allowing for detailed images of underwater objects and structures.

Limitations of Sonar Technology

1. **Limited visibility:** limited by the visibility of the water, which can be affected by factors such as sediment, algae, and water temperature. This can make it difficult to obtain clear and accurate images.
2. **Interference:** from other underwater sources, such as ships, submarines, and natural underwater features, which can lead to false readings and reduced accuracy.
3. **Short range:** Sonar signals have a limited range, which can make it difficult to image larger underwater structures or objects that are located far away from the sonar system.
4. **Limited depth:** The depth to which sonar can effectively penetrate is limited, making it unsuitable for imaging objects or structures that are located at great depths

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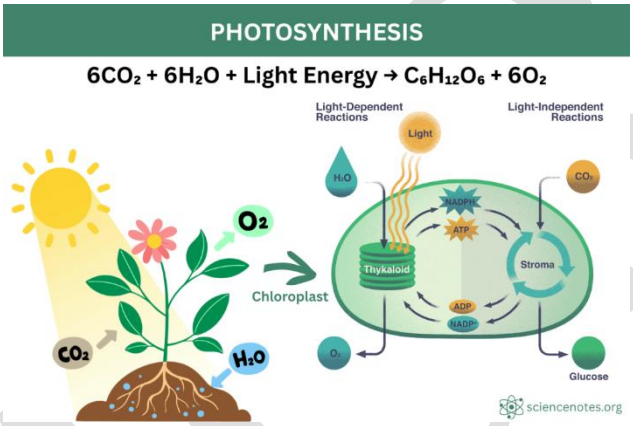
5. Acoustic noise: which can disturb marine life and harm marine ecosystems. This is particularly a concern for high-power, military-grade sonar systems, which have the potential to cause serious harm to marine life.

PHOTOSYNTHESIS:

➢ Photosynthesis is a process used by plants and other organisms to convert light energy into chemical energy.

➢ In simple terms, Photosynthesis is the process by which plants use sunlight, water, and carbon dioxide to create oxygen and energy in the form of sugar.

$$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$$



Stages of photosynthesis:

1. **Light reaction (light dependent reaction):**

a. It captures light energy to power photosynthesis.

b. Light reactions occur during the day time.

c. They take place in the thylakoids of chloroplast.

d. Pigments in the thylakoid membranes form protein complexes called Photosystem I and Photosystem II.

e. These photosystems harvest photons to produce energy carrying molecules (ATP, NADPH) that will power the dark reactions.

2. **Dark reaction (light independent reaction):**

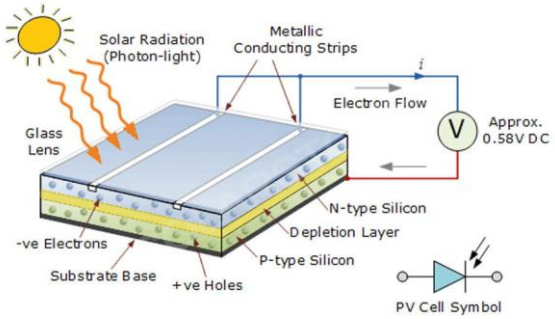
a. It does not need light energy to power their reactions and can occur day or night.

b. Dark reactions are also called Calvin Cycle.

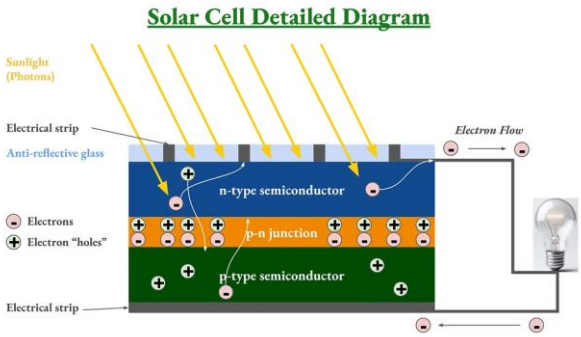
c. Dark reactions occur in the stroma of chloroplasts (the space that surrounds thylakoids) and fix carbon dioxide into glucose.

PHOTOVOLTAIC CELL:

A solar cell, or photovoltaic cell, is an electronic device that converts the energy of light directly into electricity.



Construction of a solar cell



Working:

a. Photovoltaic cells consist of two or more layers of semiconductors (silicon) with one layer containing positive charge and the other negative charge lined adjacent to each other. (p-type and n-type silicon)

b. Area around the N &P type junction, called the depletion zone, in which the electrons fill the holes.

c. Sunlight, consisting of small packets of energy termed as photons, strikes the cell, where it is either reflected, transmitted or absorbed.

d. When the photons are absorbed by the negative layer of the photovoltaic cell, the energy of the photon gets transferred to an electron in an atom of the cell.

e. With the increase in energy, the electron escapes the outer shell of the atom.

f. The freed electron naturally migrates to the positive layer creating a potential difference between the positive and the negative layer.

g. When the two layers are connected to an external circuit, the electron flows through the circuit, creating a current.

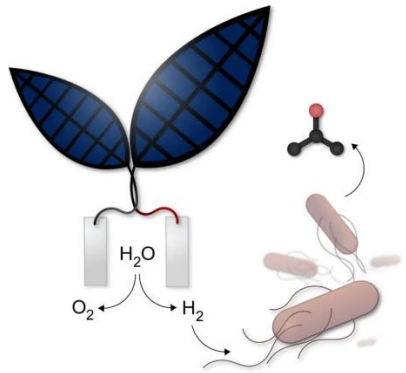
Ex: Crystelline silicon solar cell.

BIONIC LEAF

A bionic leaf is an artificial photosynthesis system that uses bacteria to convert sunlight in to usable fuel. It was first developed by researchers at Harvard University in 2015.

Components of Bionic Leaf:

a. Photosynthetic Organism
b. Light Harvesting System
c. Catalysts
d. Electron Transfer Pathway
e. Carbon Dioxide Source
f. Energy Storage or Conversion System
g. Control and Monitoring System



Working:

1. **System Composition:** The bionic leaf is typically made of a photovoltaic cell paired with a catalyst, such as a bacterium.
2. **Energy Conversion:** The photovoltaic cell captures sunlight and converts it into electrical energy, which is then passed to the catalyst.
3. **The Water Splitting Process:** Using the electrical energy, the catalyst powers a reaction that separates water molecules into hydrogen and oxygen.
4. **Role of Catalysts:** Enzymes or other catalysts serve as a bridge, facilitating the connection between the electrical energy and the chemical water-splitting reaction.
5. **Hydrogen Storage and Use:** The hydrogen produced can be stored and used as a source of energy for applications like generating electricity or powering vehicles.
6. **Environmental Benefit:** The byproduct oxygen is released into the atmosphere, helping to mitigate climate change by reducing levels of atmospheric carbon dioxide.

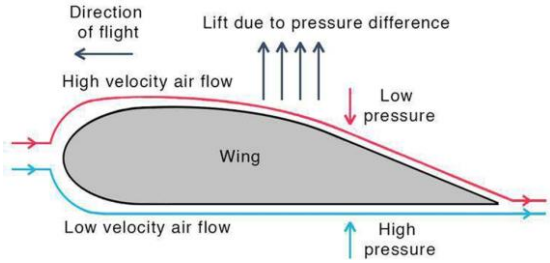
Applications of Bionic Leaf Technology:

1. Renewable Energy Production
2. Carbon Dioxide Reduction
3. Sustainable Chemical Production
4. Agriculture and Food Production
5. Remote and Off-Grid Areas
6. Environmental Remediation

BIRD FLYING:

1. Birds fly by flapping their wings and using their body weight and the movement of the air to stay aloft.
2. It seems unlikely that animals could repeat such long trips accurately if they were relying only on their sense of smell.
3. They navigate using a combination of visual cues, the Earth's magnetic field, and celestial navigation.
4. Aircraft, on the other hand, use engines to generate thrust and lift from the wings to stay in the air.

5. They navigate using a combination of instruments and systems, including GPS (Global Positioning System), which uses satellite signals to determine the aircraft's position and help it navigate.



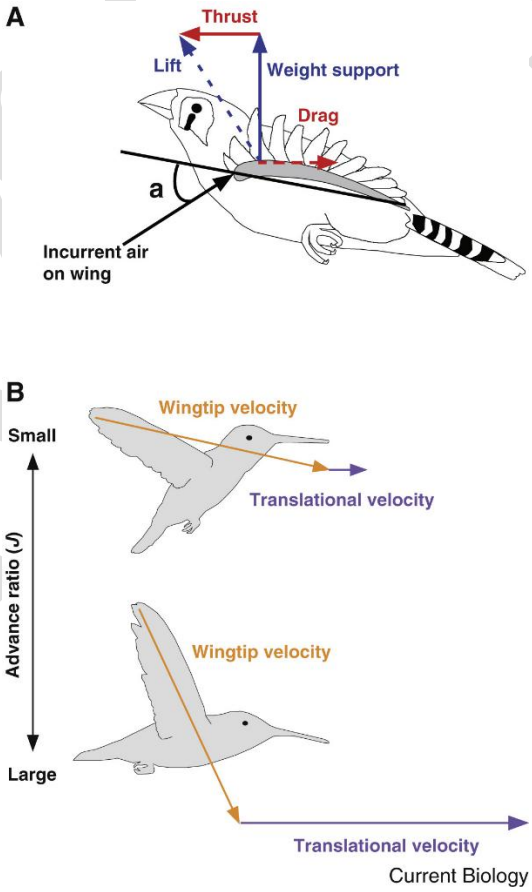
Science behind bird flying:

1. Wing Shape: The wings are curved on the upper surface and flatter on the bottom, creating a pressure difference this generates lift, allowing birds to stay airborne.

2. Wing Muscles: Birds have strong flight muscles attached to their wings, allowing them to flap their wings vigorously.

3. Hollow Bones: Birds have lightweight bones that are hollow and filled with air sacs, reducing their overall weight.

4. Feathers: They provide both lift and control.



5. Respiratory System: Birds have a unique respiratory system that allows for efficient oxygen exchange.

Bird flying, GPS (Global Positioning System)

a. Birds navigate through their inner GPS. They do not depend on GPS, instead uses celestial, magnetic & visual cues.

b. Birds developed sense of smell for navigation. Ex: Homing pigeons

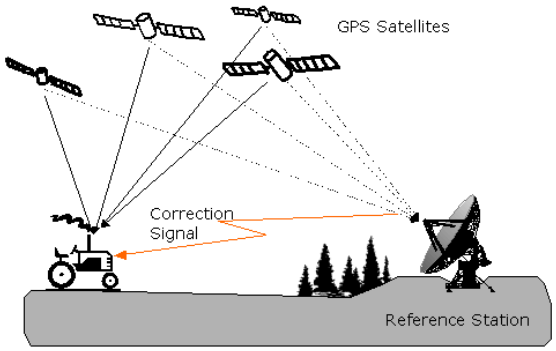
c. Homing pigeons are famous for being able to navigate extremely long distances.

d. Researchers have discovered a small spot on the beak of pigeons and some other birds that contains magnetite.

e. Magnetite is a magnetized rock, which may act as a tiny GPS unit for the homing pigeon by giving it information about its position relative to Earth's poles.

f. Researchers have also found some specialized cells in birds' eyes that may help them see magnetic fields.

g. It is thought that birds can use both the beak magnetite and the eye sensors to travel long distances over areas that do not have many landmarks, such as the ocean.



Mechanism of Bird flying in Aircraft:

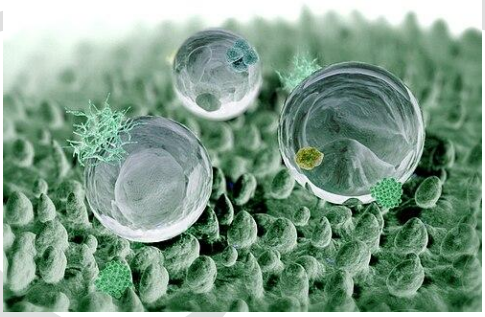
1. **Lift, drag, thrust and weight (Gravity)**
   - **a.** The fundamentals of bird flight are similar to those of aircraft, in which the aerodynamic forces sustaining flight are lift, drag, and thrust.
   - **b. Gravity,** the force that pulls everything towards earth is balanced by lift.
   - **c. Lift Generation:** Lift is produced by air flowing over the wing (airfoil), which is shaped to create a net upward force while directing air movement downward.
   - **d. Body Lift:** In some species, additional lift is generated by airflow around the bird's body, particularly during intermittent flight when wings are folded or semi-folded.
   - **e. Aerodynamic Drag:** Drag acts opposite to the direction of motion and causes energy loss. It consists of two parts:
2. **Lift-induced drag:** The unavoidable cost of producing lift.
3. **Parasitic drag:** Caused by skin friction between the air and the bird's body surfaces.
   - **a. Force Reduction:** Birds use streamlining of their bodies and wings to reduce drag forces.

b. Thrust Production: Unlike aircraft that use engines, birds generate thrust by flapping their wings at specific amplitudes and frequencies.

LOTUS LEAF EFFECT:

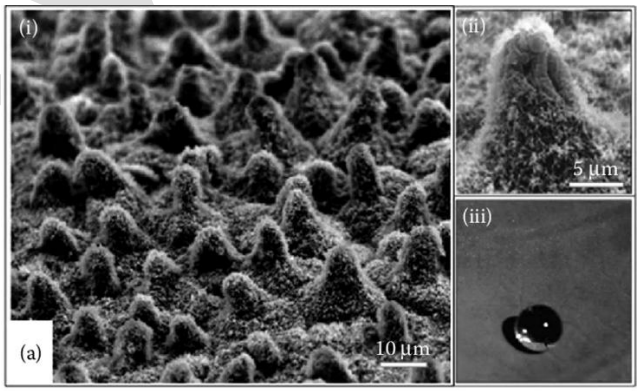
a. The ability of lotus leaves to repel water and self-clean through their unique surface structure- lotus leaf effect.
b. This effect has inspired the development of super hydrophobic and self-cleaning surfaces, which have a wide range of applications in various industries.
c. The lotus leaf surface has two properties, a microstructure and contains chemicals that consists of numerous small bumps (papillae) and wax-coated hairs.
d. This structure creates a high contact angle between the water droplets and the surface, causing the droplets to roll off and carry away any dirt or debris.
e. This self-cleaning property is due to the lotus leaf's ability to repel water and resist adhesion.

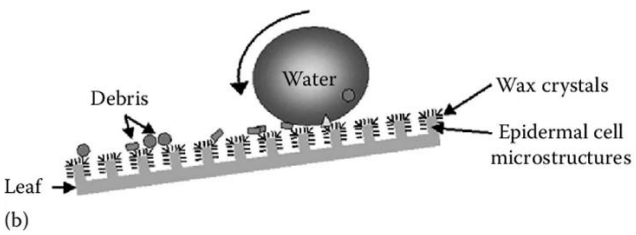




Super Hydrophobic Effect

Principle: The ability of certain surfaces to repel water and resist wetting. Super hydrophobic surfaces are characterized by a high contact angle between water droplets and the surface and a low contact angle hysteresis, meaning that the droplets roll off the surface with ease.





1. The **super hydrophobic effect** is achieved through the use of various techniques. These techniques create a surface structure that traps air between the surface and the water droplets, reducing the contact area between them and making it more difficult for the droplets to wet the surface.

2. Self-Cleaning Surfaces

It involves two main mechanisms:

a. The reduction of surface energy and the modification of surface texture.

b. These mechanisms work together to minimize the adhesion of dirt, water, and other contaminants, enabling the self-cleaning effect.

Applications:

a. Architecture and Building Materials: self-cleaning glass for windows and facades. These surfaces repel dirt, dust, and pollutants, reducing the need for frequent cleaning and maintenance.

b. Solar Panels: Self-cleaning coatings on solar panels prevent the accumulation of dust and dirt on the surface, ensuring optimal energy efficiency.

c. Automotive Industry: vehicle exteriors, including car windows and windshields.

d. Self-cleaning coatings can be used on electronic displays, touchscreens, and optical lenses.

e. Textiles: Self-cleaning coatings can be applied to fabrics used in outdoor clothing, carpets.

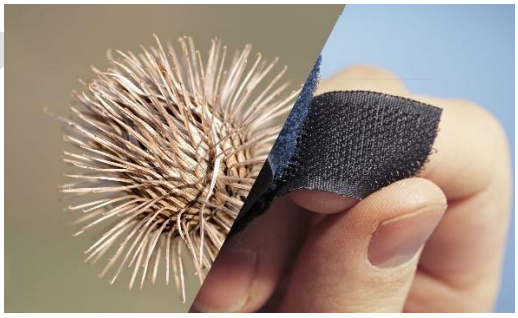
PLANT BURR:

a. A bur is a seed or dry fruit that has hooks or teeth. The main function of the bur is to spread the seeds of the bur plant, often through epizoochory.

b. The hooks of the bur are used to catch the fur or fabric, so that the bur, which contain seeds, then can be transported along with the thing it attached itself to.

c. Another use for the spines and hooks is physical protection against herbivores.

d. The stems of certain species like cactus also are covered with thorns and may function as burs.



VELCRO:

Velcro was invented by Swiss engineer George De Mestral in 1941

Working:

a. Hook and loop fasteners have two strips the rough side has hundreds of tiny 'hooks' and the softer side has hundreds of tiny "loops".

b. After pressing of two strips, the hook cling to the loops to make them stick.

c. The more hooks and loops that are attached, the stronger the hook and loop fastener will be.

d. The hooks and loops can be easily separated by pulling them apart, making Velcro a convenient way to fasten and unfasten items.

Applications:

a. Clothing and footwear: for closures and adjustable straps. It can be easily opened and closed, making it convenient for users with limited dexterity or mobility.

b. Medical devices: such as braces, splints, and compression garments for its adjustable and secure fastening capabilities.

c. Aerospace equipment: such as satellites and spacecraft, to secure components in place and prevent them from vibrating or shifting during launch or flight.

d. Automotive industry: for a range of applications, such as securing carpets and headliners, and attaching door panels and seat cushions.

e. Packaging industry: for bags, pouches, and other types of packaging.

f. Sports equipment: such as helmets and gloves, for its ability to provide a secure and adjustable fit.

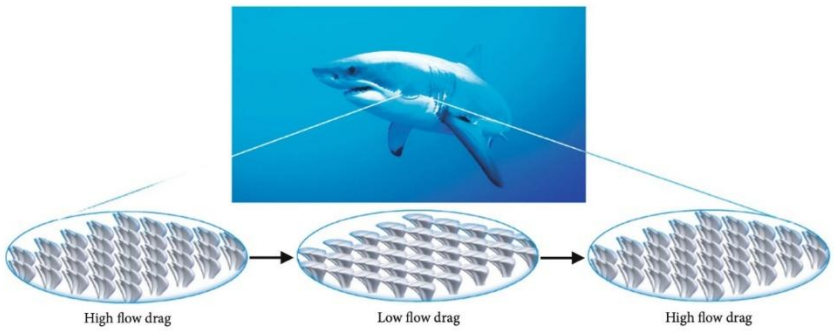
SHARK SKIN:

a. Shark skin refers to the dermal denticles that cover the body of sharks. These denticles are small, tooth-like scales that have a unique surface structure that is biomimetic

b. The shark skin is covered in tiny ridges called riblets arranged parallel that reduce the drag as the shark swims through water. Means that the shark can move more efficiently through the water.

c. It is filled with a mucus layer that helps to prevent the growth of microorganisms.

d. The shape and orientation of the scales help to minimize drag and turbulence as the shark moves through the water.



Frictionless Swim Suits

a. Which are designed to improve the performance of swimmers by reducing drag in the water.

b. Friction-reducing swim suits use a similar structure to that of shark skin to reduce drag and improve swimmer performance.

c. These suits are made from high-tech materials that mimic the properties of shark skin, such as the shape and size of the denticles.

KINGFISHER BEAK & BULLET TRAIN:

a. The kingfisher beak is an excellent example of nature's design for efficient diving and fishing.

b. Its unique shape and structure enable the kingfisher to minimize the impact of water resistance and achieve a successful dive.

c. It is much narrower and flatter than the beaks of most other birds.

d. They dive into the water with minimal splash, noise and water disruption so as to not alert the fish that they are coming.

e. The sharp beak of the kingfisher helps to pierce through the water's surface, breaking the surface tension and reducing the force required to enter the water.

f. The beak is also equipped with specialized sensors that allow them to detect the exact location of the fish. Once the fish is located, they will dive into the water with incredible speed and accuracy.



Bullet train:

a. Shinkansen trains were very loud, and although they were fast, this was an issue causing a lot of disturbance. The use of the kingfisher beak as a design inspiration for the front of the bullet train to reduce sound and increase speed.

b. The streamlined design minimizes drag as the train travels at high speeds, allowing it to maintain stability and efficiency. The smooth, tapered shape reduces the pressure difference between the front and rear of the train, reducing noise and vibration.

c. When a high-speed train moves through a tunnel, it creates pressure waves that can cause noise. The nose of the Shinkansen is designed to reduce these pressure waves by effectively managing airflow and minimizing the compression and expansion of air as the train enters and exits tunnels. This reduces the noise level.

HUMAN BLOOD SUBSTITUTE:

a. A blood substitute (also called artificial blood or blood surrogate) is a substance used to mimic and fulfill some functions of biological blood particularly oxygen transport.

b. These substitutes are designed to be used in situations where blood transfusions are not available, feasible, or when there is a risk of blood-borne infections.

Types of human blood substitutes:

1. Oxygen carriers: Oxygen carriers are designed to carry and deliver oxygen to thebody's tissues, must like hemoglobin

Ex: perfluorocarbons (synthetic molecules that can dissolve large amounts of oxygen), hemoglobin-based oxygen carriers (derived from human or animal hemoglobin

2. Plasma Expanders: Plasma expanders are designed to increase blood volume and maintain blood pressure without carrying oxygen.

Ex: Saline solutions, albumin, and hydroxyethyl starch.

3. Hemoglobin based oxygen carrier:

a. HBOC are synthetic or recombinant modules that are designed to mimic the oxygen-carrying capacity of red blood cells. They are developed as a blood substitute.

b. HBOC can be made using bovine, human or recombinant hemoglobin. The hemoglobin is then purified & modified to increase stability & reduce toxicity.

c. Some of the modifications include cross-linking between hemoglobin molecules or conjugating hemoglobin to a polymer.

d. Once it is fused into bloodstream, the HBOCs dissolve & release hemoglobin, which binds with oxygen in the lungs & transports it to the tissue that need it.

e. Unlike RBC, HBOCs do not have a flexible membrane and cannot circulate for long periods in the bloodstream. They are typically used as a temporary measure until the patient can receive a blood transfusion.

f. Have been used in clinical trials for a variety of medical conditions such as trauma, surgery and anaemia. Potential to cause kidney damage, increase the risk of heart attack or stroke, & interfere with blood tests.

Ex: Hemopure, Oxyglobin

Advantages:

a. Increased oxygen-carrying capacity
b. Universal compatibility
c. Longer shelf life
d. Reduced risk of infections
e. Availability in remote or challenging settings

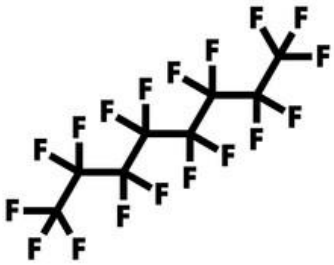
Limitations:

a. Limited oxygen release
b. Short half-life
c. Nitric oxide scavenging.
d. Renal toxicity
e. Immunogenicity and adverse reactions

Perfluorocarbon

1. PFCs are a group of synthetic chemicals that are chemically stable, non-toxic, and have the ability to dissolve and transport oxygen.
2. Unlike HBOCs, PFCs do not bind to oxygen instead physically dissolve it. So, it does not couses vasocontraction.
3. When PFCs are infused into the bloodstream, they are taken up by RBCs and transport oxygen from lungs to body tissues.
4. Ex: Perftoran, Oxycyte

Structure of PFCs



perfluorooctane

Advantages:

a. High oxygen-carrying capacity
b. Improved oxygen solubility
c. Stability and long shelf life
d. No blood typing or cross-matching required
e. Reduced risk of infection transmission

Limitations:

1. Limited oxygen offloading
2. Short half-life
3. Side effects and toxicity
4. Need for specialized administration methods
5. Clearance and elimination

MCR